PROJECT REPORT

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

1. INTRODUCTION

1.1 Project Overview

The **Auto Ticket Classification Using Flow Designer** project is designed to automate the classification of IT support incident requests in a school environment. The IT helpdesk receives different types of requests from students and teachers, such as Wi-Fi connectivity problems, projector failures, password issues, and slow computer performance.

In the existing process, IT support staff manually read each incident, understand the reported problem, and select the appropriate category and subcategory. When the number of incidents increases, this process becomes repetitive and time-consuming.

The proposed system uses **ServiceNow Flow Designer** to automate this classification process. When a new incident is created, the system analyzes the **Short Description** and **Description** fields using predefined keywords and conditions. Based on the identified keywords, the incident is automatically assigned to the appropriate category and subcategory.

The system reduces manual effort and provides a consistent method for classifying common IT incidents. It also allows the classification rules to be modified when new types of IT issues are introduced.

1.2 Purpose

The main purpose of the project is to develop an automated incident classification process that reduces the manual effort required by IT support staff.

The objectives are:

- To automatically classify newly created IT incidents.
- To analyze the Short Description and Description of an incident.
- To identify relevant keywords from the incident details.
- To assign the appropriate category and subcategory.
- To reduce repetitive manual classification work.
- To improve consistency in ticket classification.
- To help IT support staff process incidents more efficiently.
- To provide a simple and maintainable automation process.

2. IDEATION PHASE

2.1 Problem Statement

The school IT helpdesk receives a large number of incident requests from students and teachers. These requests include different problems such as Wi-Fi connectivity issues, projector failures, password problems, and slow computers.

Currently, IT support staff manually review every incident and select the appropriate category. This process takes time and increases the workload of the support team. Manual classification can also result in inconsistent categorization or incorrect classification.

Problem Statement:

As an IT support team member, I need an automated way to classify incoming incident requests based on their descriptions so that common issues can be categorized quickly and consistently, reducing manual effort and improving ticket-processing efficiency.

2.2 Empathy Map Canvas

The empathy map focuses on the **IT support team member**, who is responsible for handling incident requests.

Says

- "Manual classification takes too much time."
- "I need a faster way to categorize tickets."
- "Tickets should reach the correct support team."
- "Automation can reduce repetitive work."

Thinks

- Wants to process tickets quickly and accurately.
- Is concerned about incorrect classification.
- Wants to reduce repetitive work.
- Wants a simple and reliable process.

Does

- Receives incident requests.
- Reads the Short Description and Description.
- Identifies the issue type.
- Selects the category and subcategory.

- Tracks and resolves incidents.

Hears

- Requests to assign incidents correctly.
- Complaints about delayed issue resolution.
- Feedback about incorrectly categorized tickets.
- Requests to reduce ticket-processing time.

Pains

- Manual classification is time-consuming.
- Large numbers of incidents increase workload.
- Incorrect categorization can occur.
- Repetitive work reduces efficiency.

Gains

- Automatic classification.
- Reduced manual effort.
- Consistent categorization.
- Faster ticket processing.
- Easier incident management.

2.3 Brainstorming

During the brainstorming stage, the team discussed common problems in IT service management and identified manual incident classification as an important problem.

Several possible ideas were considered:

1. Automatic incident classification.
2. Keyword-based ticket categorization.
3. Automatic category and subcategory assignment.
4. Automated ticket routing.
5. Automatic notifications.
6. Ticket status updates.
7. Classification history.

8. Predefined classification rules.

After evaluating the ideas based on feasibility, usefulness, and project requirements, **Auto Ticket Classification Using Flow Designer** was selected as the proposed solution.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey describes the experience of a student or teacher while submitting an IT incident.

Stage 1 – Create Incident

The user submits an IT-related issue through ServiceNow.

Stage 2 – Enter Information

The user provides information in the Short Description and Description fields.

Stage 3 – Process Incident

The newly created incident triggers the Flow Designer workflow.

Stage 4 – Classify Ticket

The system analyzes predefined keywords and conditions and assigns the appropriate category and subcategory.

Stage 5 – Notification

The IT support team is notified about the classified incident when notification is configured.

Stage 6 – Resolve

The IT support team reviews the classified incident and works on the reported issue.

Stage 7 – Close

After resolution, the incident is updated and closed.

3.2 Solution Requirements

Functional Requirements

FR No.	Requirement
FR-1	The system shall allow users to create IT incident requests.

FR No.	Requirement
FR-2	The system shall accept Short Description and Description details.
FR-3	The system shall trigger Flow Designer when a new incident is created.
FR-4	The system shall analyze incident information using predefined keywords and conditions.
FR-5	The system shall automatically identify the appropriate issue category.
FR-6	The system shall assign the appropriate subcategory.
FR-7	The system shall update the incident record with the classification result.
FR-8	The system shall allow classification rules to be modified or extended.

Non-Functional Requirements

Usability

The system should be simple for IT support staff to use and maintain.

Security

Incident information and system configurations should be accessible only to authorized users.

Reliability

The workflow should consistently apply the configured classification rules.

Performance

The system should process newly created incidents efficiently.

Availability

The classification workflow should be available whenever the ServiceNow instance is available.

Scalability

New keywords, categories, subcategories, and conditions should be added when required.

3.3 Data Flow Diagram

The Data Flow Diagram represents the flow of incident information within the system.

Students and teachers submit IT-related incident requests through ServiceNow. The incident information, including the Short Description and Description, is stored in the incident record. When a new incident is created, Flow Designer is automatically triggered.

The workflow reads the incident information and analyzes it using predefined keywords and conditions. Based on the identified issue, the system determines the appropriate category and subcategory and updates the incident record.

The classified incident is then available to the IT support team for investigation and resolution.

Main Data Flow:

Student/Teacher → Incident Submission → Incident Record → Flow Designer → Keyword Analysis → Automatic Classification → Category/Subcategory Update → IT Support Team

3.4 Technology Stack

Component	Technology
User Interface	ServiceNow
Incident Management	ServiceNow Incident Table
Automation	ServiceNow Flow Designer
Classification	Predefined Keyword-Based Conditions
Data Storage	ServiceNow Platform Database
Notifications	ServiceNow Notifications
Infrastructure	ServiceNow Cloud Platform
Access Control	ServiceNow Roles and Permissions
Machine Learning	Not used

The project uses a **rule-based classification approach** rather than a machine-learning model. This makes the solution simple to configure and maintain.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The school IT helpdesk faces difficulties because every incident must be manually reviewed and categorized. The proposed solution addresses this problem by using Flow Designer to automatically analyze incident information.

The system connects the helpdesk requirement for faster and consistent classification with an automated workflow. Predefined keywords and conditions are used to identify common issues such as Wi-Fi, projector, password, and slow computer problems.

The solution reduces repetitive work and allows IT support staff to spend more time on technical issue resolution rather than manual ticket categorization.

4.2 Proposed Solution

The proposed **Auto Ticket Classification Using Flow Designer** system automatically classifies IT incident requests.

When an incident is created, Flow Designer is triggered. It reads the Short Description and Description fields and checks them against predefined keywords and conditions.

For example:

- Wi-Fi or network-related keywords → Network/Wi-Fi category
- Projector-related keywords → Projector category
- Password-related keywords → Password category
- Slow computer-related keywords → Computer/Performance category

After identifying the issue, the system automatically updates the appropriate category and subcategory in the incident record.

This approach reduces manual classification, improves consistency, and provides a maintainable solution for the IT helpdesk.

4.3 Solution Architecture

The system consists of the following major components:

1. **Student/Teacher** – submits the IT incident.
2. **ServiceNow Incident Interface** – receives the incident request.
3. **Incident Record** – stores the Short Description and Description.
4. **Flow Designer** – triggers when the incident is created.
5. **Keyword and Condition Analysis** – checks the incident information against predefined rules.

6. **Automatic Classification** – determines the appropriate category and subcategory.
7. **Incident Record Update** – stores the classification result.
8. **IT Support Team** – reviews and resolves the classified incident.

Architecture

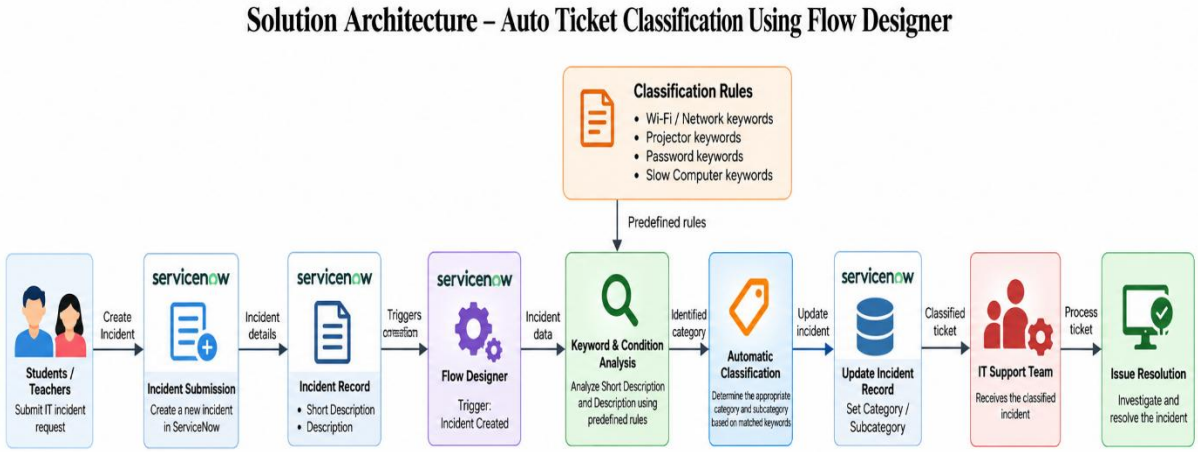


Figure 1: Architecture and data flow of the Auto Ticket Classification Using Flow Designer system

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is planned using Agile concepts such as **Epics, User Stories, Story Points, Sprints, Velocity, and Burndown Chart**.

Sprint Planning

Sprint	Main Activity	Story Points
Sprint-1	Requirement Analysis & Planning	9
Sprint-2	Backend Development & Configuration	12
Sprint-3	Automation Using Flow Designer	15
Sprint-4	Testing & Validation	11
Sprint-5	Deployment & Documentation	7
Total		**54**

Velocity

Total Story Points = 54

Number of Sprints = 5

Average Velocity = 54 / 5 = 10.8 Story Points per Sprint

Therefore, the team's average velocity is **10.8 story points per sprint**.

Sprint Schedule

Sprint	Start Date	End Date
Sprint-1	01 Oct 2026	06 Oct 2026
Sprint-2	08 Oct 2026	13 Oct 2026
Sprint-3	15 Oct 2026	20 Oct 2026
Sprint-4	22 Oct 2026	27 Oct 2026
Sprint-5	29 Oct 2026	03 Nov 2026

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing was conducted to verify that the Flow Designer workflow processes incident records efficiently.

The following scenarios were tested:

- Flow trigger after incident creation.
- Processing of Short Description and Description.
- Keyword identification.
- Automatic classification.
- Category and subcategory update.
- Processing of multiple incident records.
- Consistency of classification results.

The workflow successfully processed the tested incidents according to the configured rules.

Performance Testing Result: PASS

6.2 Functional Testing

Functional testing verified whether each major function of the system worked as expected.

Test	Expected Result	Result
Create incident	Incident record created	Pass
Enter Short Description	Information stored	Pass
Enter Description	Information stored	Pass
Trigger Flow Designer	Flow starts automatically	Pass
Analyze keywords	Relevant keywords identified	Pass
Classify ticket	Correct category assigned	Pass
Update subcategory	Correct subcategory assigned	Pass
Verify incident	Classification displayed	Pass

6.3 User Acceptance Testing

UAT was performed using common IT incidents.

Test Case	Scenario	Result
UAT-01	Wi-Fi issue	Pass
UAT-02	Projector issue	Pass
UAT-03	Password issue	Pass
UAT-04	Slow computer issue	Pass
UAT-05	Verify category/subcategory	Pass
UAT-06	Verify Flow Designer trigger	Pass

Total Test Cases: 6
Passed: 6
Failed: 0

UAT Pass Percentage = 100%

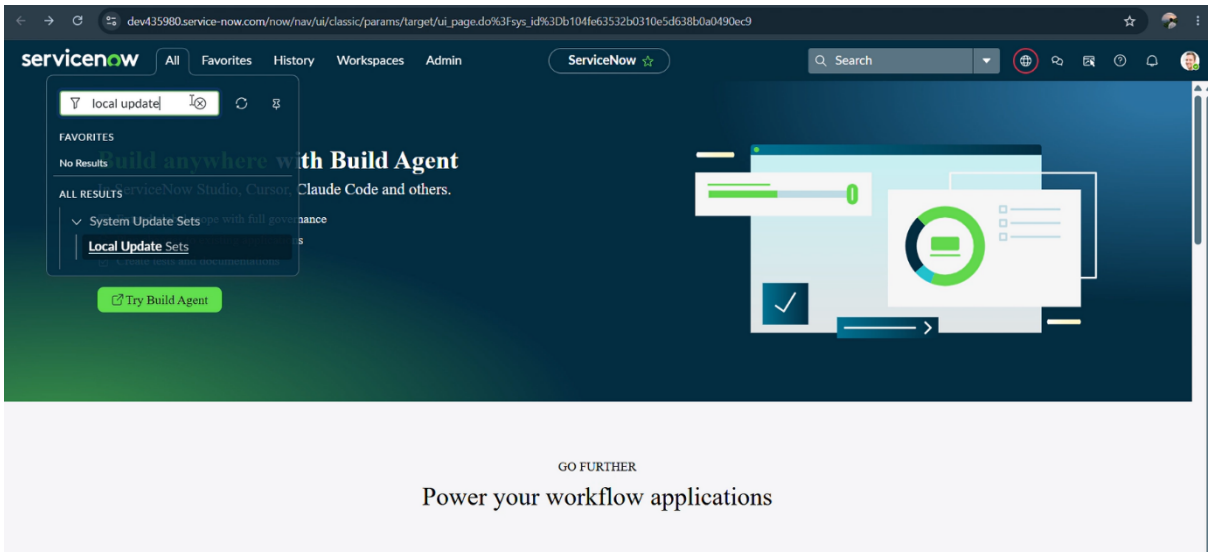
The system was accepted for the tested requirements.

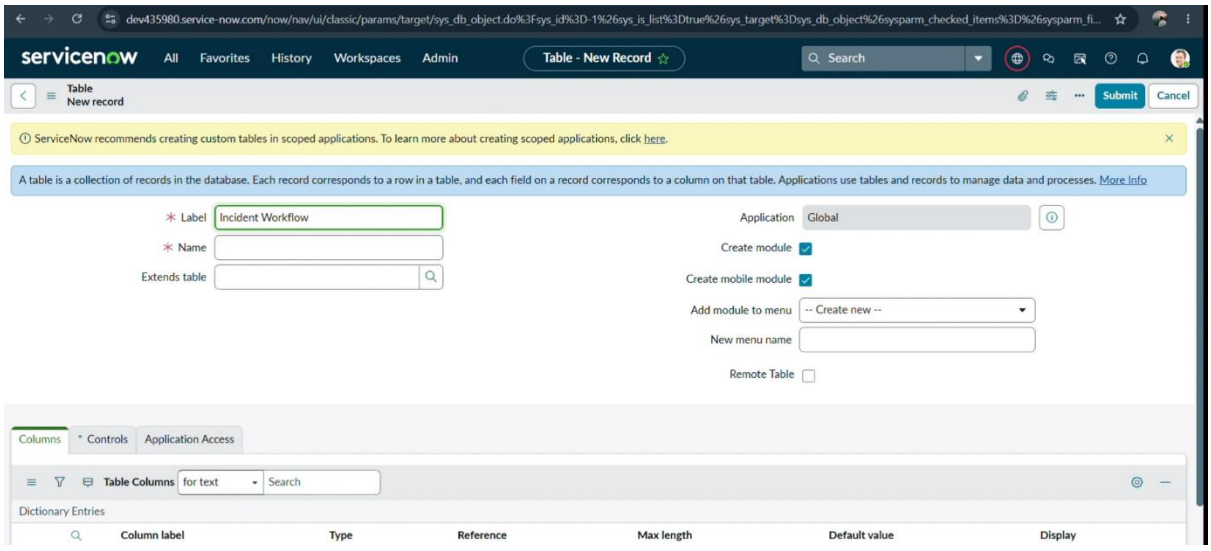
7. RESULTS

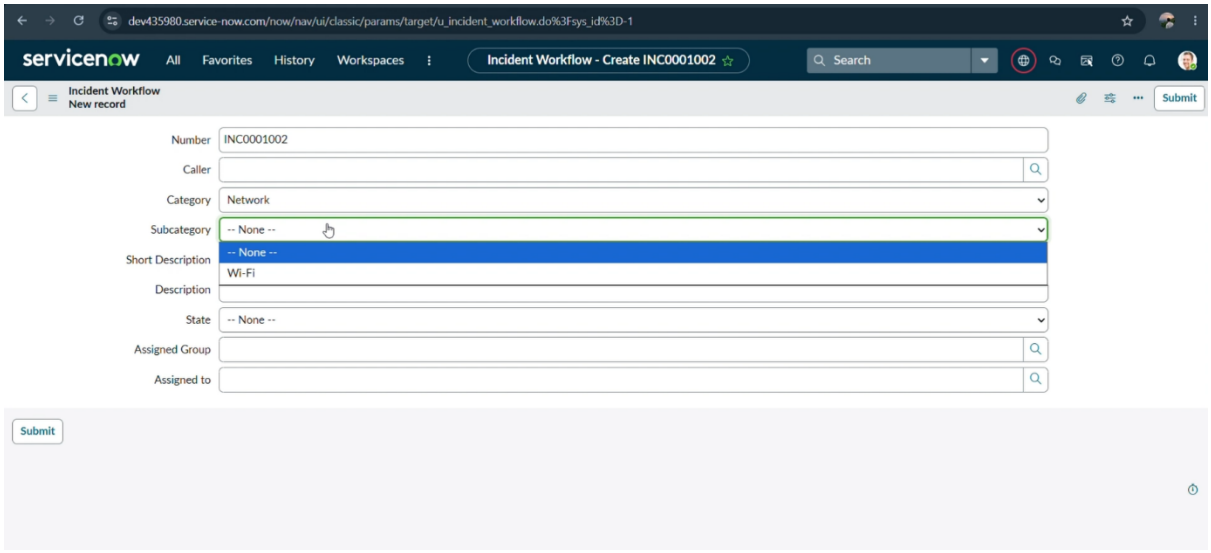
7.1 Output Screenshots

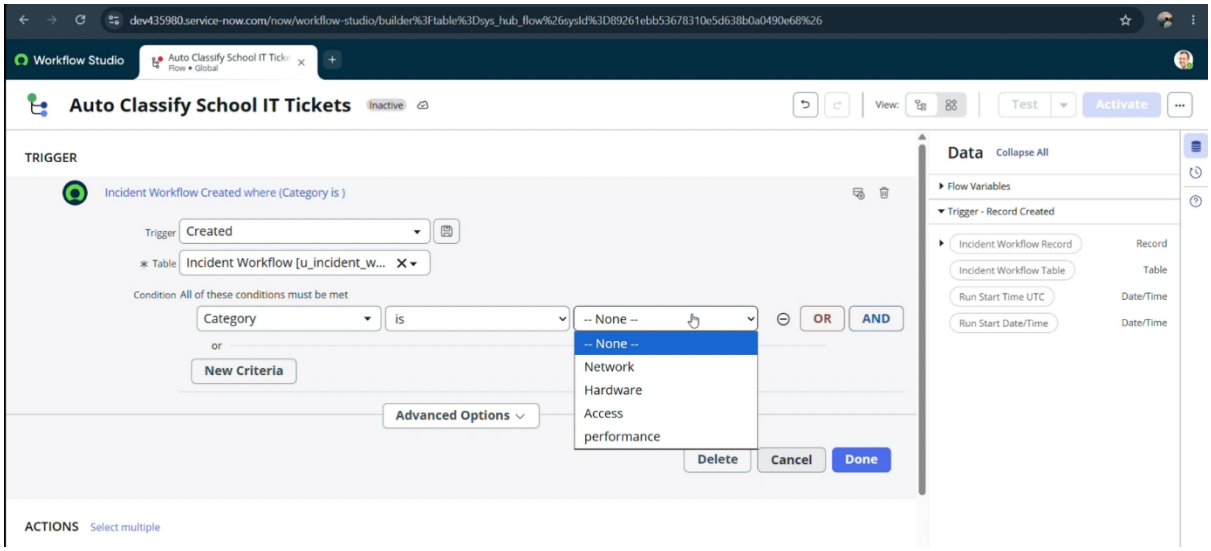
The following screenshots should be included in the project report:

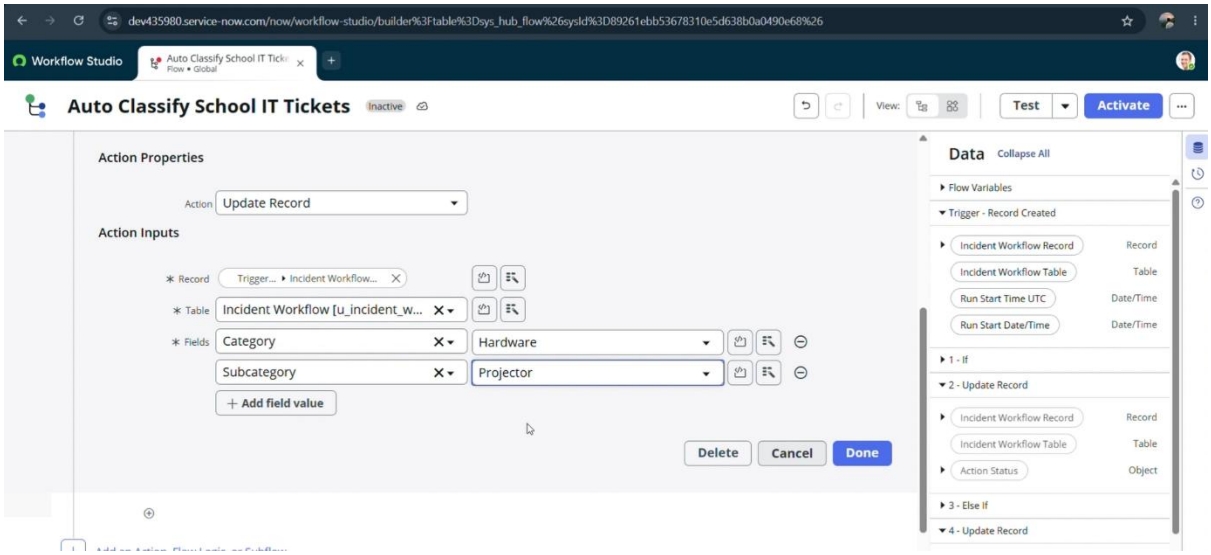
ServiceNow

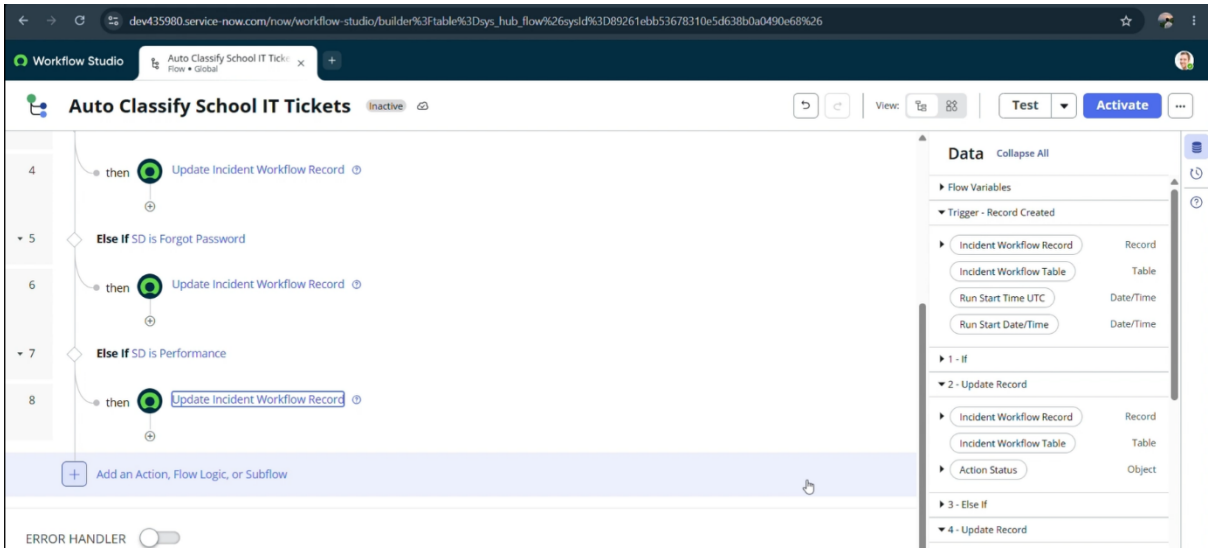




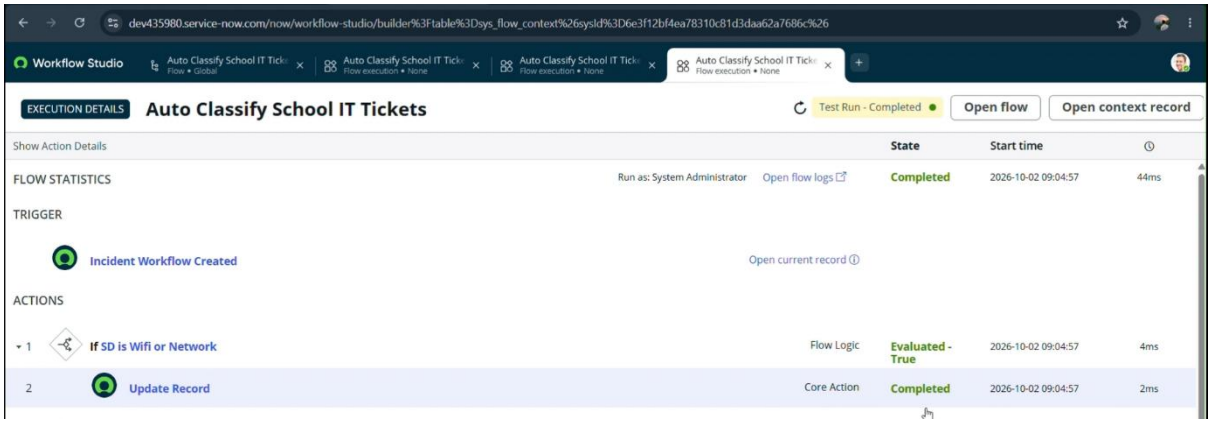

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servicenow
All
Favorites
History
Workspaces
Incident Workflow - Create INC0001002
Search
Incident Workflow
New record
Submit
Number
INC0001002
Caller
Category
Network
Subcategory
-- None --
-- None --
Wi-Fi
Short Description
Description
State
-- None --
Assigned Group
Assigned to
Submit

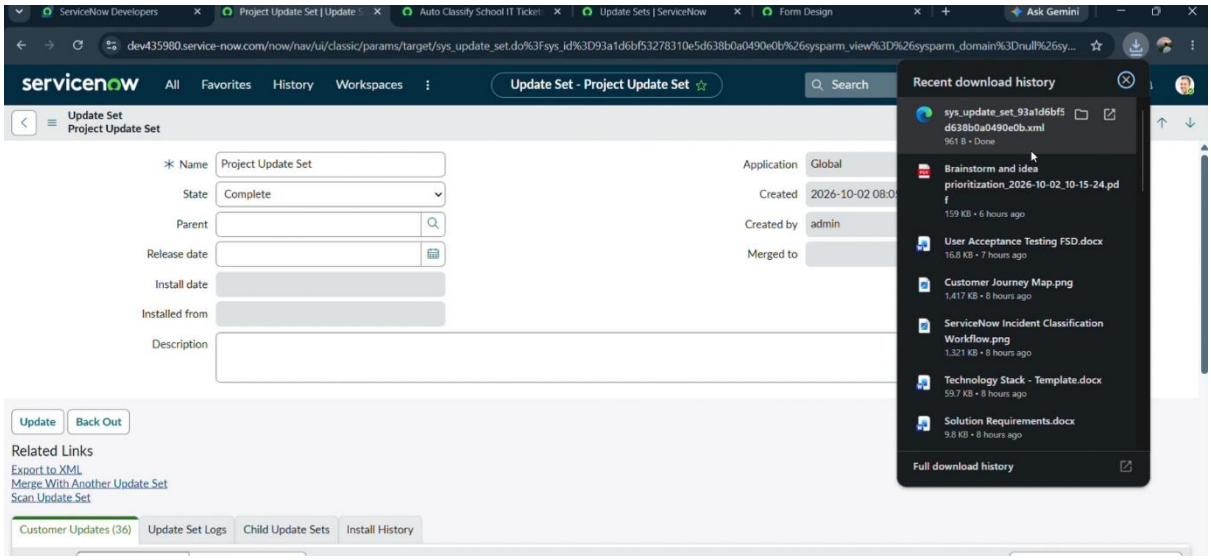

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Workflow Studio
Auto Classify School IT Tick
Flow • Global
Auto Classify School IT Tickets
Inactive
View:
Test
Activate
TRIGGER
Incident Workflow Created where (Category is)
Trigger
Created
Table
Incident Workflow [u_incident_w...
Condition
All of these conditions must be met
Category
is
-- None --
OR
AND
or
New Criteria
-- None --
Network
Hardware
Access
performance
Advanced Options
Delete
Cancel
Done
ACTIONS
Select multiple
Data
Collapse All
Flow Variables
Trigger - Record Created
Incident Workflow Record
Record
Incident Workflow Table
Table
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Date/Time
Run Start Date/Time
Date/Time


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Incident Workflow [u_incident_w...
Fields
Category
Hardware
Subcategory
Projector
Add field value
Delete
Cancel
Done
Data
Collapse All
Flow Variables
Trigger - Record Created
Incident Workflow Record
Record
Incident Workflow Table
Table
Run Start Time UTC
Date/Time
Run Start Date/Time
Date/Time
1 - If
2 - Update Record
Incident Workflow Record
Record
Incident Workflow Table
Table
Action Status
Object
3 - Else If
4 - Update Record



Testing Result





8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Reduces manual ticket classification.
- Saves IT support staff time.
- Provides consistent classification.
- Automatically processes newly created incidents.
- Easy to maintain and configure.
- Supports common IT issue categories.
- Reduces repetitive work.
- Can be extended with additional keywords and conditions.

8.2 Disadvantages

- Classification depends on predefined keywords and rules.
- Unrecognized keywords may not be classified correctly.
- Poorly written incident descriptions may affect classification.
- New issue types require additional classification rules.
- The solution depends on the availability of the ServiceNow platform.

9. CONCLUSION

The **Auto Ticket Classification Using Flow Designer** project provides an automated approach for classifying IT support incidents in a school environment. The system uses ServiceNow Flow Designer to analyze the Short Description and Description of newly created incidents and apply predefined keyword-based conditions.

The solution successfully classifies common incidents such as Wi-Fi problems, projector failures, password issues, and slow computer problems. The automatic assignment of categories and subcategories reduces the need for repetitive manual classification by IT support staff.

The testing results demonstrate that the configured workflow performs the required classification functions for the tested scenarios. The project provides a simple, maintainable, and scalable approach for improving the efficiency of IT incident management.

10. FUTURE SCOPE

The system can be enhanced in the future with additional capabilities such as:

1. Adding more IT issue categories and classification keywords.
2. Supporting more complex classification conditions.
3. Integrating automated ticket routing based on the classification.
4. Adding priority assignment based on incident information.
5. Providing dashboards for monitoring incident classifications.
6. Adding feedback mechanisms to improve classification rules.
7. Integrating advanced text-analysis techniques for more complex incident descriptions.
8. Extending the solution to other departments and organizations.
9. Adding multilingual incident classification.
10. Integrating knowledge-base suggestions for frequently reported issues.

11. APPENDIX

11.1 Source Code / Configuration Details

The project primarily uses **ServiceNow Flow Designer configuration** rather than conventional application source code.

The main configuration includes:

- Incident table configuration.
- Incident fields.
- Flow Designer trigger.
- Keyword-based conditions.
- Category and subcategory actions.
- Notification configuration.
- Testing and validation rules.

11.2 Dataset Link

Not Applicable.

The project does not use a machine-learning dataset. It uses predefined keywords and conditions for rule-based ticket classification.

11.3 GitHub Link

https://github.com/naveen-01-07/-Auto-Ticket-Classification-using-Flow-Designer

11.4 Project Demo Link

https://drive.google.com/file/d/1TgAgGgBy9BC0sIplPjrfDdMUVdHnDdoC/view?usp=sharing

11.5 Reference

https://www.servicenow.com/in/products/platform-flow-designer.html